

Business Problem Statement

A mid-sized retail company wants to better understand its customers' shopping behaviour in order to improve sales, customer retention, and loyalty program effectiveness. The management team has noticed variation in purchasing patterns across age groups, genders, product categories, and seasons, and wants to understand how factors such as subscription status, promo code usage, payment method, and review ratings relate to customer spending and repeat purchase behaviour.

You are tasked with analyzing the company's customer shopping dataset to answer the following overarching business question:

"How can the company use its customer shopping data to identify high-value customer segments, understand the drivers of repeat purchases and subscriptions, and inform targeted marketing and loyalty strategies?"

Deliverables

1. **Data Preparation (Python):** Clean and transform the raw customer shopping dataset — handling missing values, removing redundant fields, standardizing column names, and engineering new features (e.g. purchase frequency in days, age groups) to prepare the data for structured analysis.
2. **Data Analysis (SQL):** Load the cleaned data into SQL Server and write structured queries across four key business themes — Customer Overview, Product Category Performance, Marketing & Promotions, and Customer Loyalty — to uncover patterns in revenue, subscription behaviour, and repeat purchases.
3. **Visualization & Insights (Power BI):** Build a multi-page interactive dashboard that highlights key customer segments, product trends, and loyalty patterns, enabling stakeholders to explore the data and make informed decisions.
4. **Report and Presentation:** Summarize key findings — such as which segments drive the most revenue, what influences subscription uptake, and which categories underperform — along with actionable business recommendations.
5. **GitHub Repository:** Publish all Python notebooks, SQL scripts, and the Power BI dashboard file in a well-structured, documented repository.